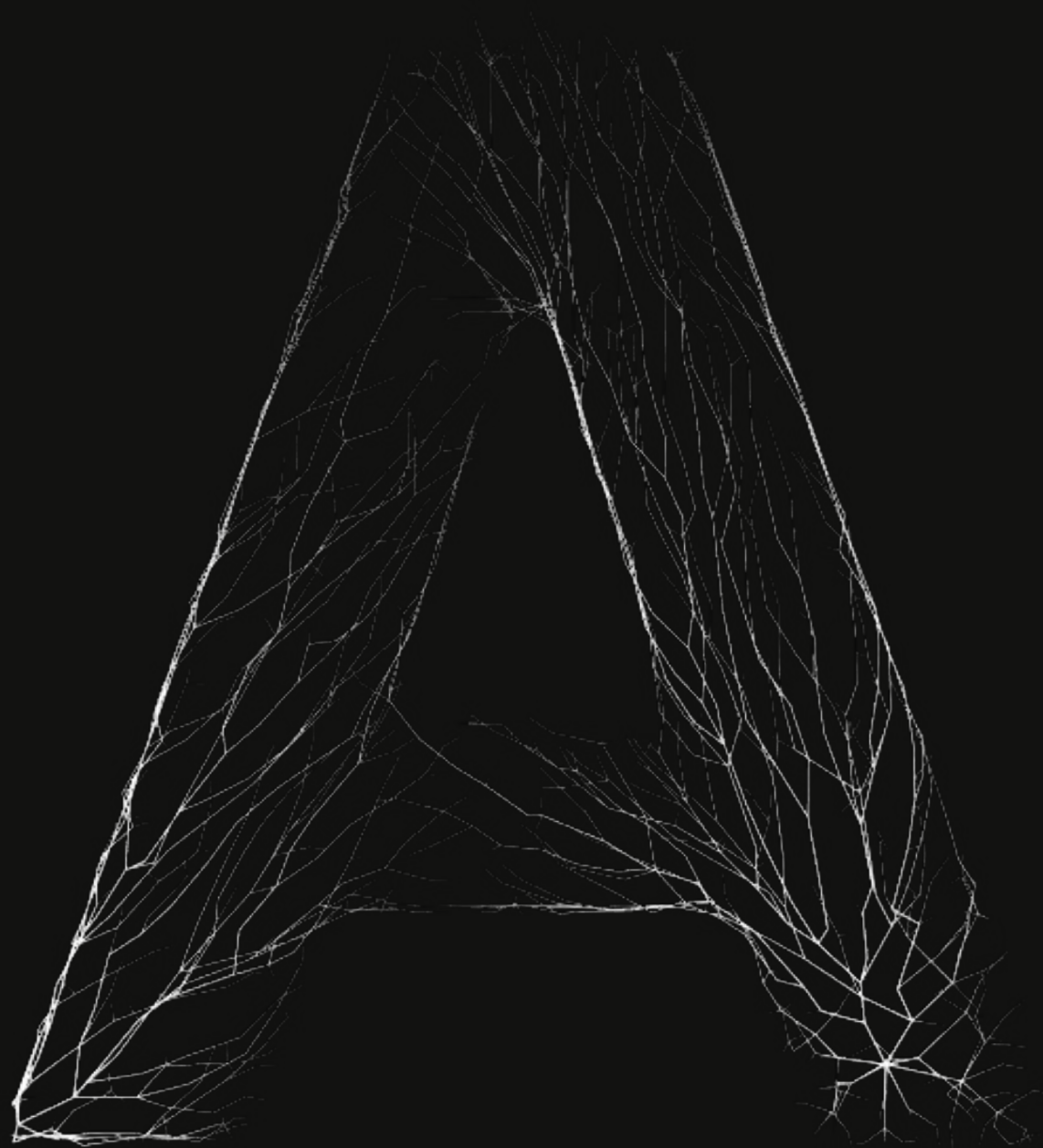




RETHINKING .LOGISTICS

Adam Večeřa

KURILLA - PROKOP ATELIER

ARCHIP - ARCHITECTURAL INSTITUTE IN PRAGUE • SUMMER SEMESTER 2025-26



"I don't believe in staying in one lane. To me, architecture is the ultimate puzzle because it demands everything at once - logic, art, and grit.

My background is a mix of high-speed interests. I've designed layouts for magazines, captured stories through a camera lens, and pushed my physical limits in bodybuilding. Each of these fields has taught me something vital about composition, discipline, and the way human beings occupy and live in the space.

I'm a student who looks for the 'connective tissue' between different industries, and I'm at my best when I'm staring down a challenge that others might find intimidating.

I applied this exact mindset to the RETHINKING.LOGISTICS studio, where I thrived on the complexity of navigating a massive, difficult-to-solve topic and turning it into a tangible architectural vision."

KURILLA - PROKOP ATELIER

REGENERATIVE LOGISTICS

CONSUMER-INTEGRATED STORAGE AND DISTRIBUTION WAREHOUSE • IKEA ČERNÝ MOST, PRAHA 14

Adam Večeřa

ARCHIP - ARCHITECTURAL INSTITUTE IN PRAGUE • SUMMER SEMESTER 2025-26

Regenerative

Parametric

Consumer integrated

Contents

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WHAT IS REGENERATIVE LOGISTICS?

From extraction to regeneration

Regenerative logistics moves beyond sustainability — beyond doing less harm — to actively restoring ecological, social and economic systems. The building is not a neutral container but a productive urban actor.

Traditional logistics infrastructure extracts value from its surroundings. Regenerative logistics gives back: habitat, public space, clean energy, circular materials, community.

KEY PRINCIPLES

Ecology positive

Net biodiversity gain · habitat creation · ecological corridors

Energy positive

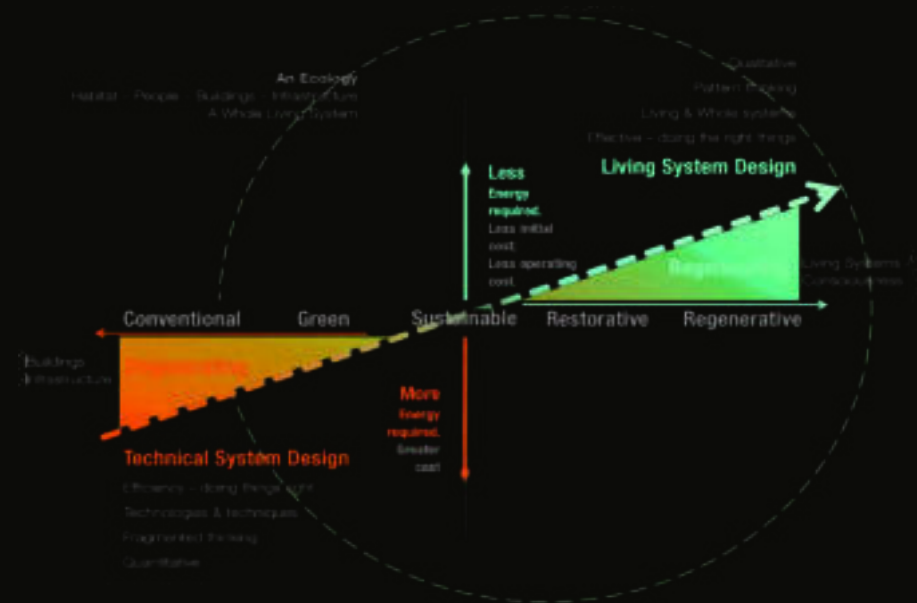
On-site renewable generation exceeds consumption

Circular by design

Closed-loop water · recycled materials · zero waste

Socially integrated

Public space · mixed program · active ground floor



THE REGENERATIVE SPECTRUM

STAGE 01 — EXTRACTIVE

Conventional warehouse

Maximises floor area · ignores context · car-dependent · landscape destroyed

STAGE 02 — SUSTAINABLE

BREEAM / LEED certified

Reduces harm · some green landscaping · still mono-use · still a barrier

STAGE 03 — RESTORATIVE

Net-zero + mixed use

Gives back what it takes · habitat created · public space · closed-loop systems

STAGE 04 — REGENERATIVE

REGENERATIVE.LOGISTICS

Actively improves urban ecology · more biodiversity · more public space · city node not barrier

30 Key Performance Indicators across 6 categories

Each KPI is measurable, target-driven, and directly informs spatial design decisions. Together they form a complete regenerative performance framework.

KPI 1

Ecology & landscape

KPI 1.1 — Biodiversity net gain +35%

KPI 1.2 — Habitat units 1→6

KPI 1.3 — Tree canopy 62%

KPI 1.4 — Green space 8.4m²/visitor

KPI 1.5 — Ecological corridor E–W

5

KPIS

KPI 2

Energy & carbon

KPI 2.1 — EUI 85 kWh/m²/yr

KPI 2.2 — Operational carbon ≤10 kgCO₂e

KPI 2.3 — Embodied ≤350 kgCO₂e/m²

KPI 2.4 — Renewable ≥90%

KPI 2.5 — Airtightness ≤1.0 q50

5

KPIS

KPI 3

Water & climate

KPI 3.1 — Water reuse 60%

KPI 3.2 — Retention 800m³

KPI 3.3 — Permeable ≥60%

KPI 3.4 — Stormwater ≥50% peak

KPI 3.5 — UHI -3°C

5

KPIS

KPI 4

Urban & mobility

KPI 4.1 — Walk score 75

KPI 4.2 — Transit 5min metro

KPI 4.3 — 300 bike spaces

KPI 4.4 — Mixed-use 6 functions

KPI 4.5 — Public space 45%

5

KPIS

KPI 5

Indoor environment

KPI 5.1 — Daylight ≥65% DA

KPI 5.2 — CO₂ ≤800ppm

KPI 5.3 — Thermal ≥90% satisfied

KPI 5.4 — Acoustic zone-specific

KPI 5.5 — Satisfaction ≥80%

5

KPIS

KPI 6

Materials & circular

KPI 6.1 — Materialepyramiden

KPI 6.2 — Recycled ≥50%

KPI 6.3 — Waste diverted ≥90%

KPI 6.4 — Reuse 40%

KPI 6.5 — Life-cycle cost -60% OpEx

5

KPIS

HORIZONTAL → VERTICAL TRANSFORMATION

EXISTING

Horizontal box

~400×400m · mono-use · car-dependent

→

PROPOSED

Vertical tower

~120 000 m² freed · public · green

BUILDING PROGRAM — STACKED LAYERS

■ Roof — Green canopy + solar + biodiversity	—
■ L7–10 — Planning studios + flexible	variable
■ L4–6 — Showrooms · living · bedroom · kitchen	12 000 m²
■ L3 — Market hall + checkout	9 000 m²
■ L2 — Restaurant + bistro + food market	3 750 m²
■ GF+L1 — Lobby + entrance + outdoor park	5 000 m²
■ B1 — High-bay automated storage	84 000 m²
■ B2 — Underground parking	1 200 spaces

~65m

height

1 200

parking sp.

84 000m²

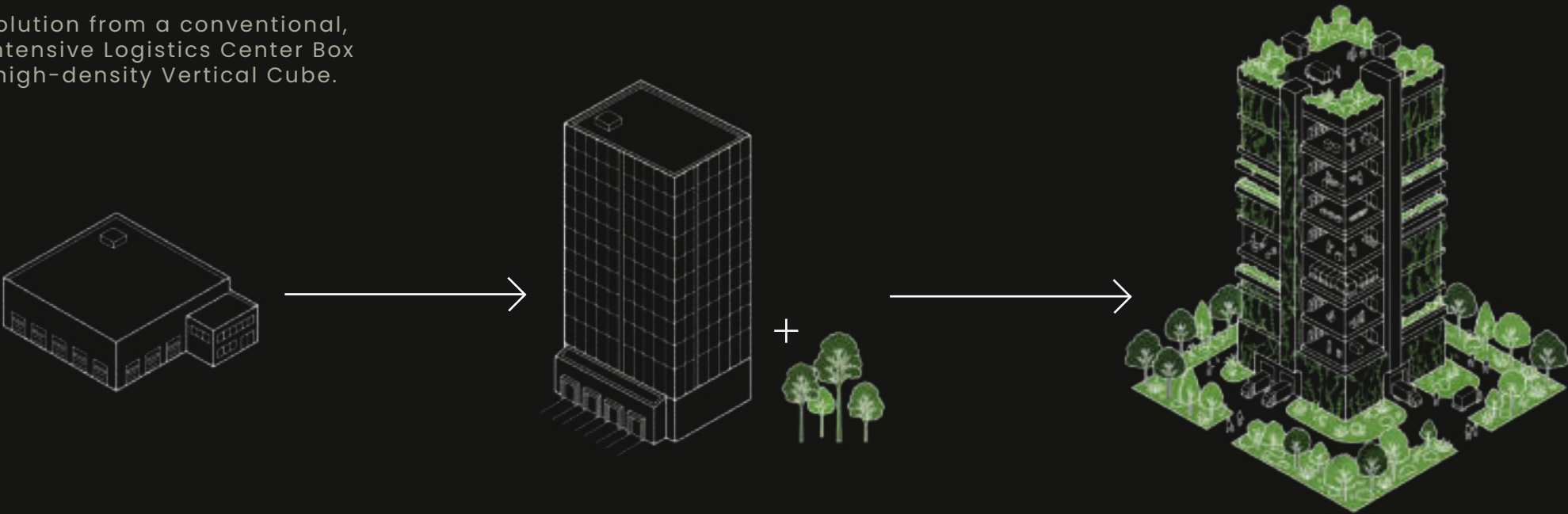
storage

10–14

floors

CONCEPT DIAGRAM

The evolution from a conventional, land-intensive Logistics Center Box into a high-density Vertical Cube.



PARAMETRIC SYSTEM — GRASSHOPPER / RHINO

Three parametric modules generate the building configuration for any city site. All dimensions adjustable — the system produces a different building for each location while maintaining the same regenerative logic.

MODULE 01

Underground parking

1 200 spaces · 6m floor · structural grid 8×8m · ramp access from street

MODULE 02

High-bay automated storage

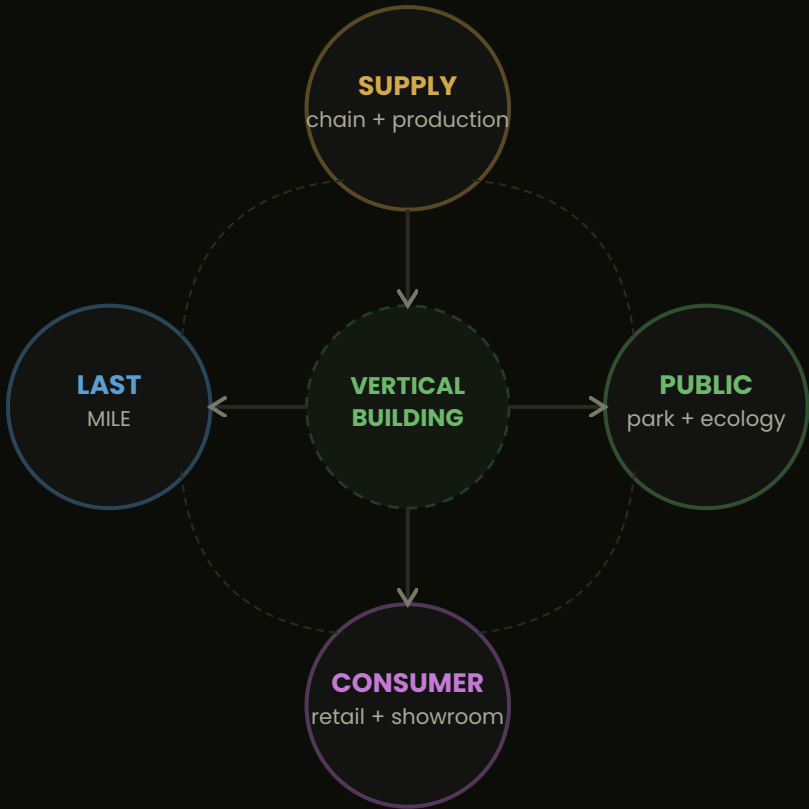
84 000 m² · 12m floor height · truck access north · stacker crane rails

MODULE 03

Mixed-use public floors

Lobby · showrooms · market · restaurants · studios · parametric area allocation

CIRCULAR BUSINESS MODEL — FOUR NODES



TRADITIONAL VS MERGED MODEL

Traditional logistics

Warehouse → trucks → distribution → shops → customer drives. Four separate trips, four facilities, no public value generated.

Merged regenerative model

Storage, showroom, market and public park share one vertical structure at a metro node. One trip, one building, maximum urban value.

REVENUE STREAMS

Logistics lease

IKEA pays per m² for automated storage

Retail + F&B

Market hall, restaurant, bistro generate independent revenue

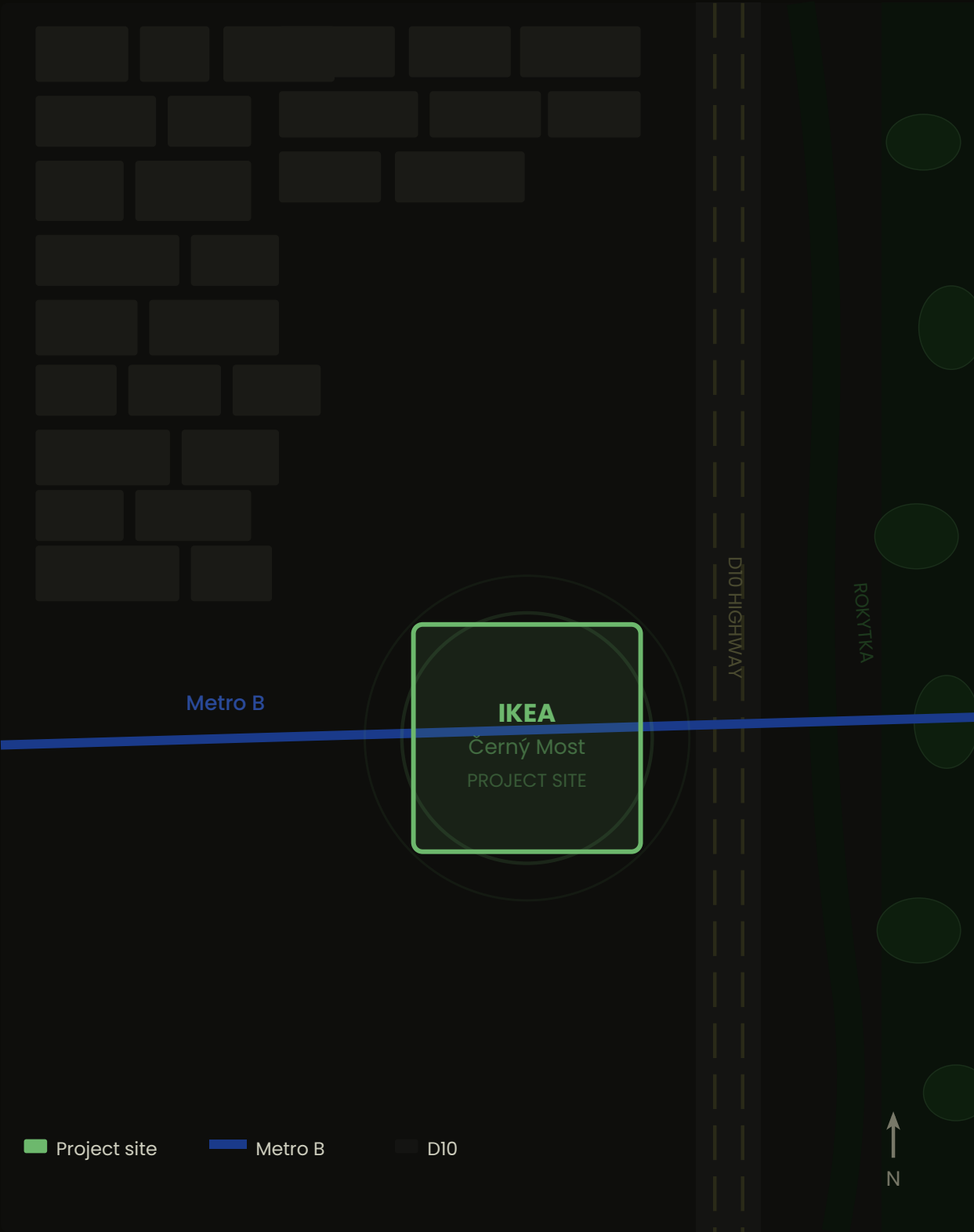
Planning studios

Flexible workspace rental for architects browsing showrooms

Energy sale

Surplus renewable energy sold back to grid from PV roof

PRAGUE CONTEXT — PRAHA 14 · ČERNÝ MOST



SITE ADDRESS

Chlumecká 664/10, Praha 14

Urban edge — where residential Černý Most meets the D10 highway and the Rokytka ecological valley.

THREE-BAND CONTEXT

West — residential

Panel housing · Metro B Černý Most · Centrum Černý Most · pedestrian paths

Centre — project site

IKEA Černý Most · surface parking · horizontal retail · poor pedestrian connectivity

East — infrastructure + nature

D10 highway · Pražský okruh · Rokytka valley · light industrial beyond

Metro B

terminus 1km

D10

direct access

Rokytka

valley 200m E

Praha 14

suburban edge

Car access

9/10

D10 + PRAŽSKÝ OKRUH · DIRECT HIGHWAY



Strong car access from D10 and Pražský okruh. Currently dominant mode. Project maintains access while reducing surface parking.

Bike access

4 → 8/10

ROUTE A26 + ROKYTKA CYCLE PATH



Currently poor — no dedicated cycle infrastructure. Proposed A26 connection will transform cycling accessibility to 8/10 after intervention.

Transit access

7/10

METRO B TERMINUS · BUS HUB

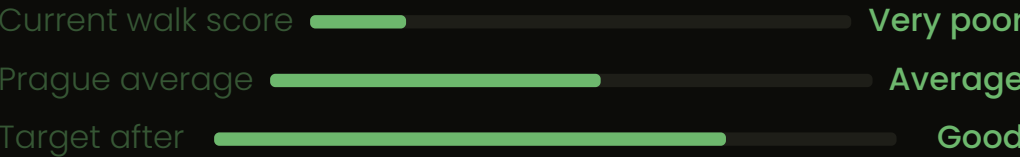


Metro B Černý Most terminus is 1km away. Proposed covered walkway would reduce to 280m effective distance — 5 min walk.

Walk access

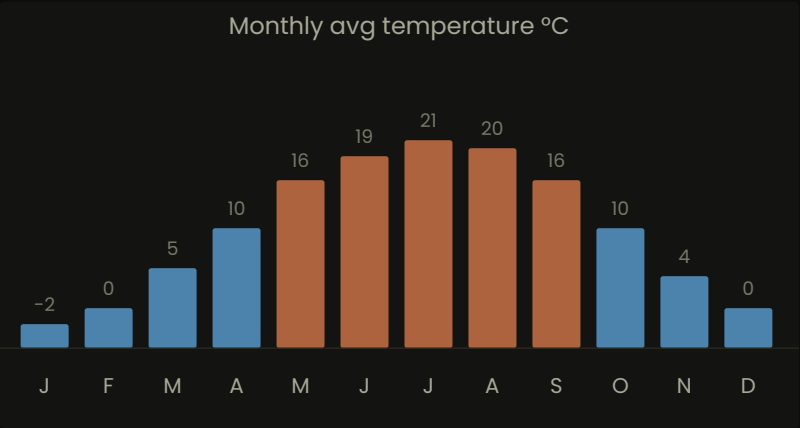
3 → 7/10

CURRENT POOR · TARGET 75 WALK SCORE

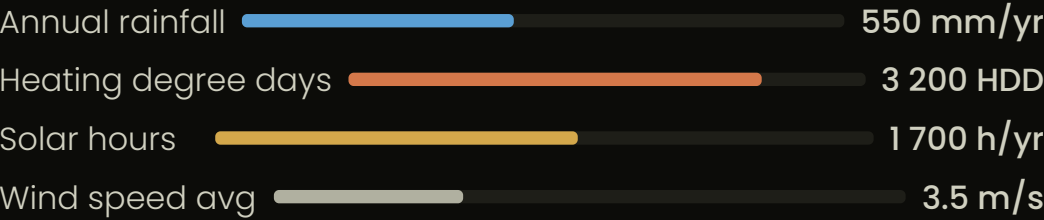


Current site is hostile to pedestrians — dominated by surface parking. New public ground floor and park would transform walkability to 75/100.

PRAGUE CLIMATE DATA — PRAHA 14



KEY CLIMATE FACTS

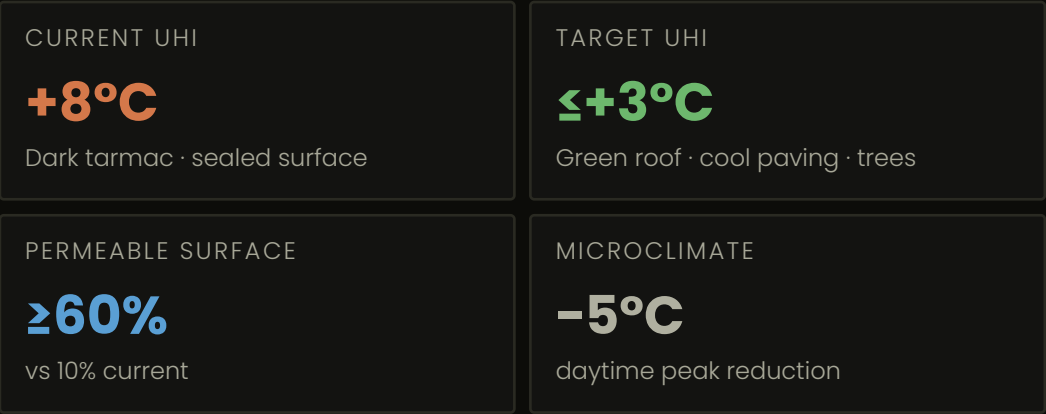


BIODIVERSITY BASELINE — EXISTING SPECIES

Current IKEA site is predominantly sealed surface with minimal ecological value. Survey identified ruderal species in neglected margins only.

- Common reed (Phragmites australis) Riparian edge
- Black elder (Sambucus nigra) Ruderal shrub
- Common knotgrass (Polygonum aviculare) Ground cover
- White clover (Trifolium repens) Ground cover
- Broad-leaved dock (Rumex obtusifolius) Ruderal
- Target: 24+ native species after Proposed planting

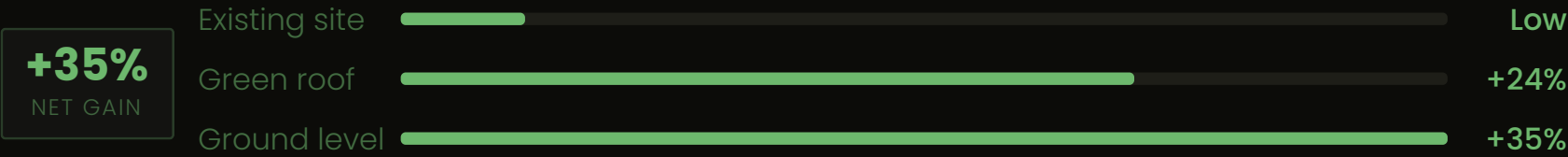
URBAN HEAT ISLAND — CURRENT VS PROPOSED



KPI 1.1

Biodiversity net gain

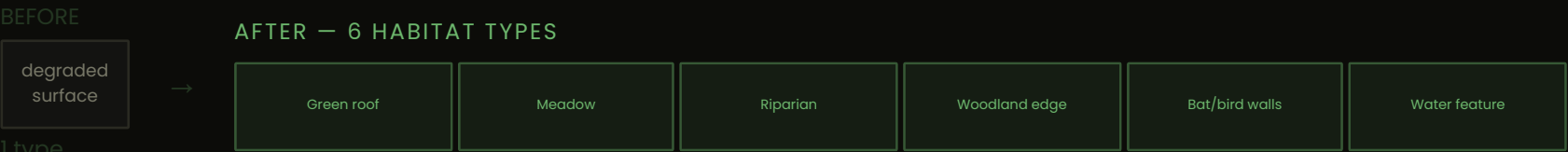
% net gain in biodiversity units vs baseline



KPI 1.2

Habitat units

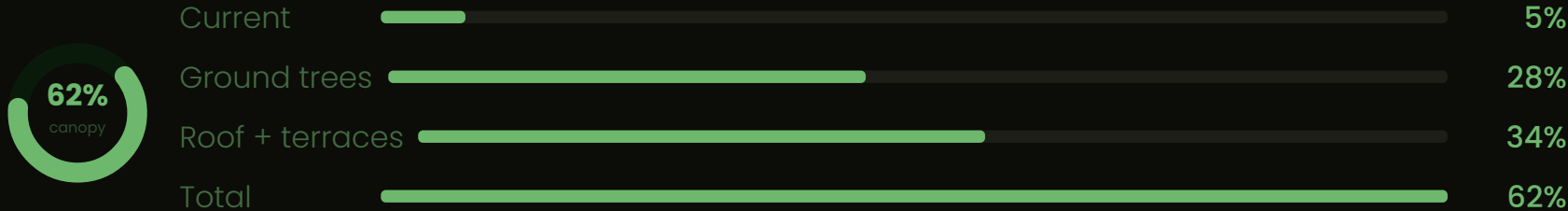
Number of distinct habitat types on site



KPI 1.3

Tree canopy coverage

% of site footprint shaded by tree canopy



KPI 1.4

Green space per capita

m² of accessible green space per daily visitor



KPI 1.5

Ecological connectivity

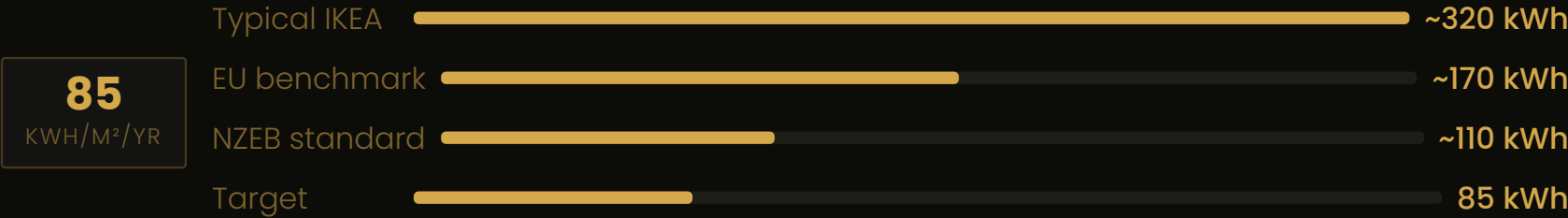
E–W corridor connecting residential to Rokytka valley



KPI 2.1

Energy use intensity

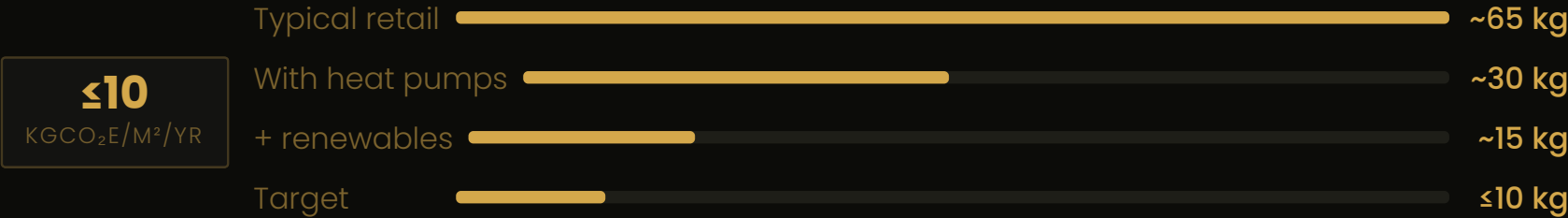
kWh/m²/yr — total energy consumed per unit of floor area



KPI 2.2

Operational carbon

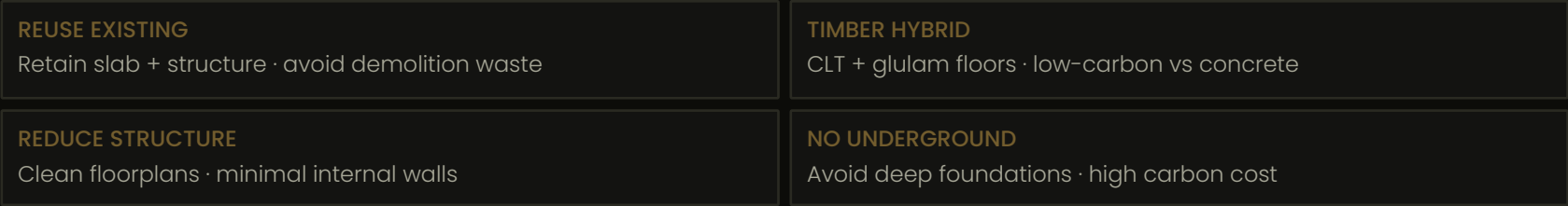
kgCO₂e/m²/yr from building use — heating, cooling, lighting



KPI 2.3

Embodied carbon

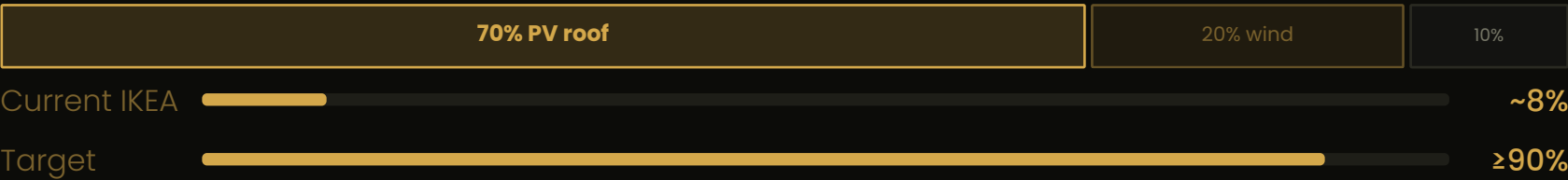
kgCO₂e/m² lifecycle stages A1–A3



KPI 2.4

Renewable energy share

% of total energy demand met by on-site renewable sources



KPI 2.5

Building airtightness

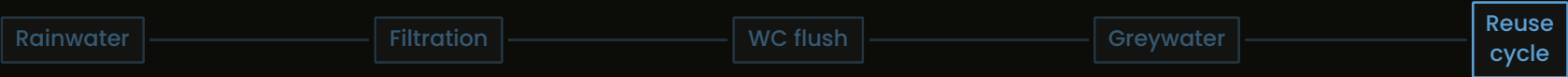
Air leakage rate q50 (m³/h·m²) — lower = better insulation



KPI 3.1

Water consumption & reuse

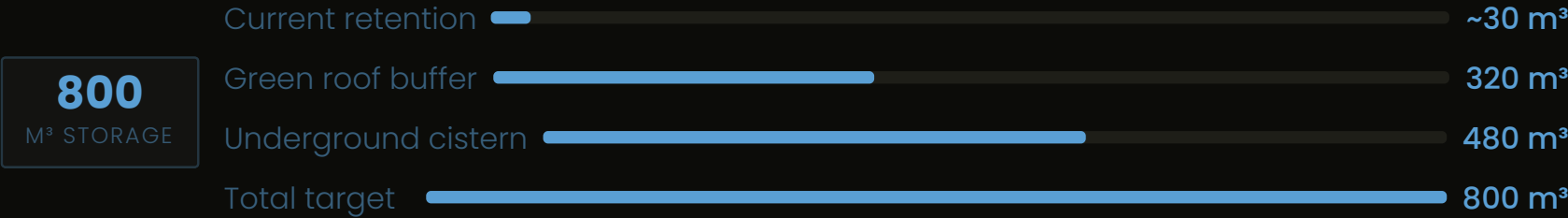
Closed-loop water system — greywater and rainwater recycled on site



KPI 3.2

Rainwater retention

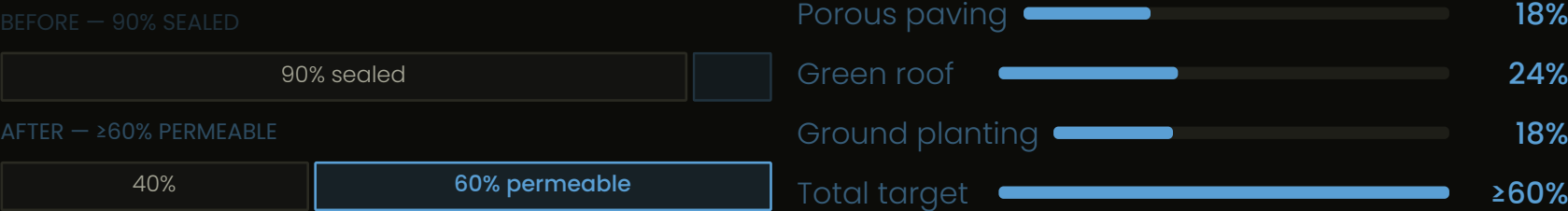
Volume of rainwater retained on site before entering drainage



KPI 3.3

Permeable surface ratio

% of site area allowing water infiltration into the ground



KPI 3.4

Stormwater infiltration

Managing peak runoff through on-site infiltration systems

BIOSWALES

Vegetated channels slow + filter runoff

RAIN GARDENS

Planted depressions absorb peak events

GREEN ROOF DELAY

Substrate holds 40–60mm before releasing

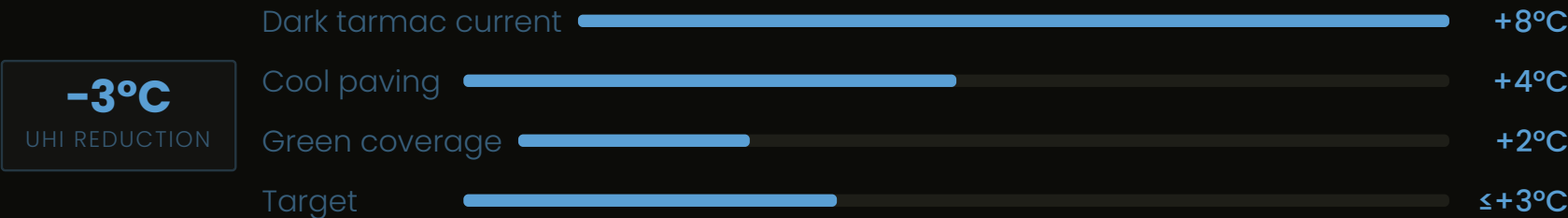
TARGET

≥50% peak flow reduction vs current

KPI 3.5

Urban heat island reduction

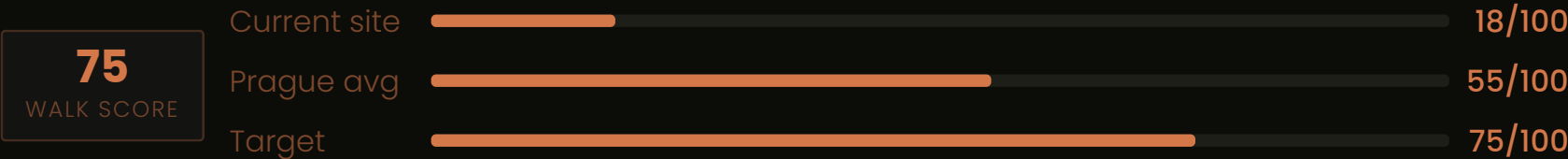
Reducing surface temperatures via vegetation + cool surfaces



KPI 4.1

Walkability index

Walk score — quality of pedestrian environment



KPI 4.2

Distance to transit

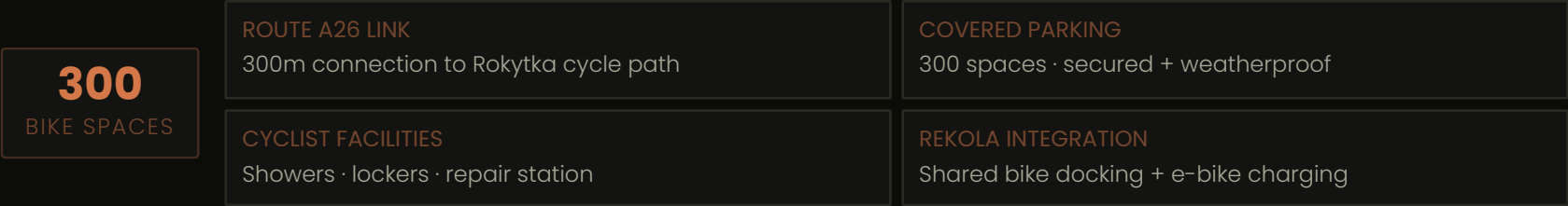
Walking time from building entrance to nearest stop



KPI 4.3

Cycling infrastructure

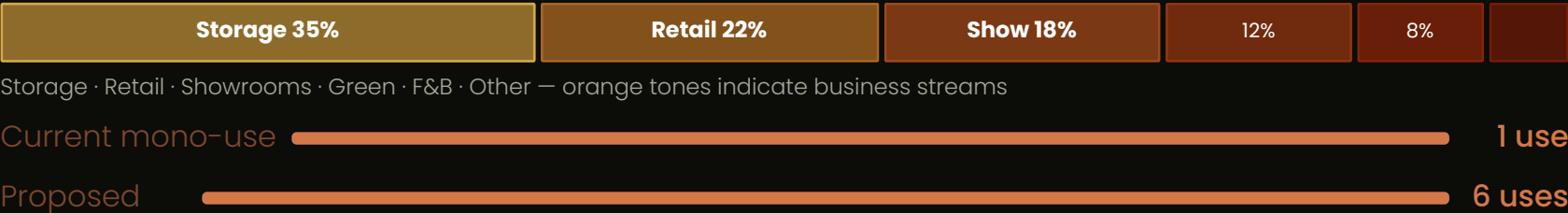
Dedicated bike routes, parking and facilities



KPI 4.4

Mixed-use ratio

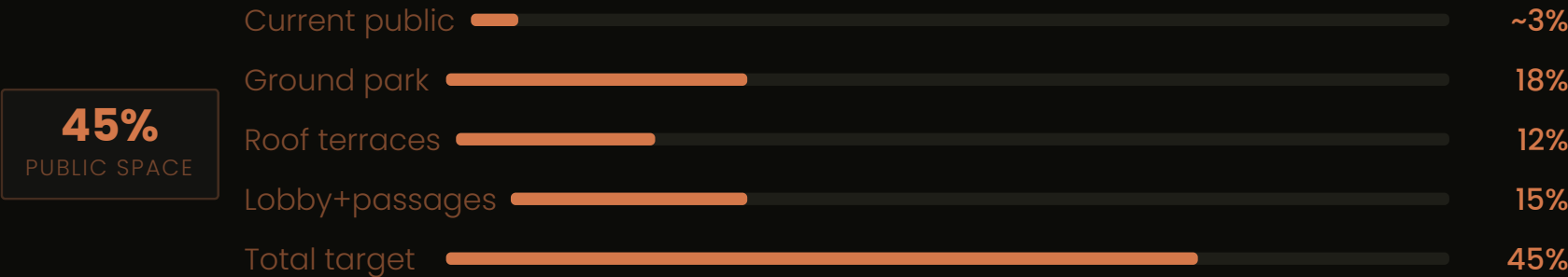
Distribution of program functions across the building — orange tones



KPI 4.5

Public space coverage

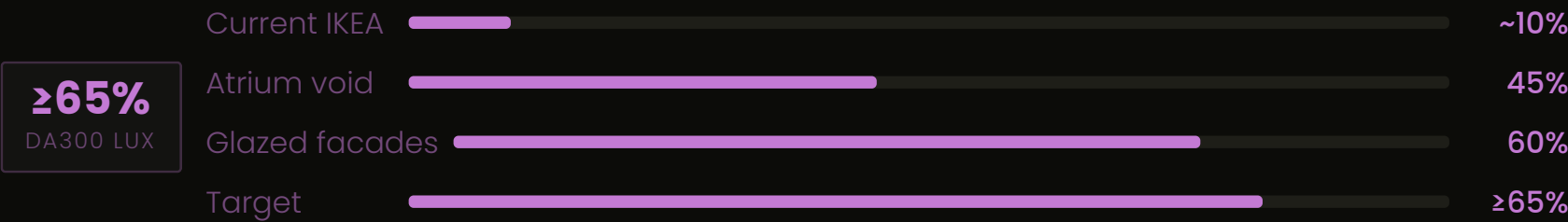
% of site area freely accessible as public space



KPI 5.1

Daylight autonomy

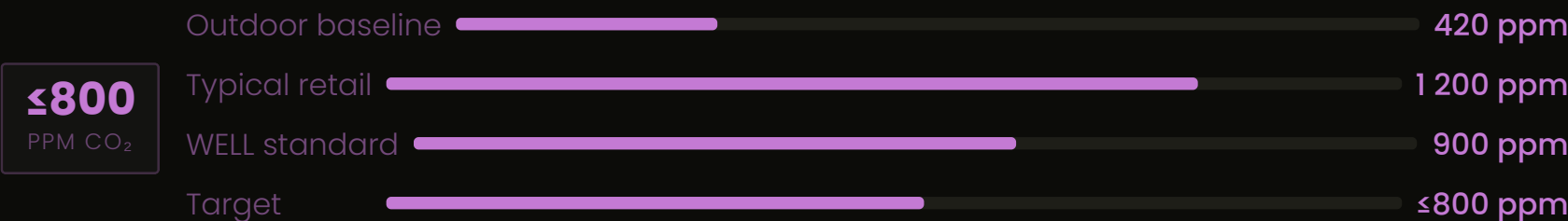
% of occupied hours met by daylight — DA300 lux



KPI 5.2

Indoor CO₂ concentration

ppm CO₂ — indicator of ventilation and air quality



KPI 5.3

Thermal comfort

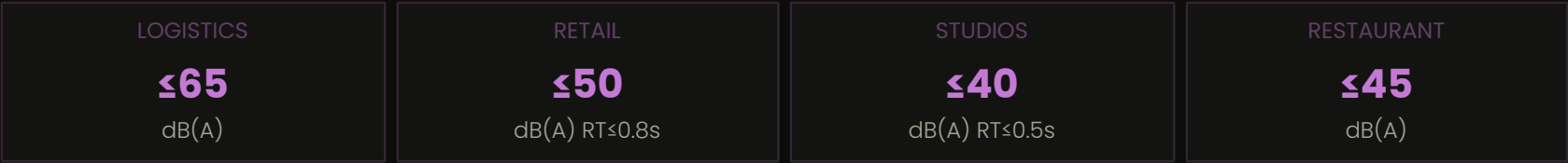
% occupants in comfort zone — EN 16798 / ASHRAE 55



KPI 5.4

Acoustic comfort

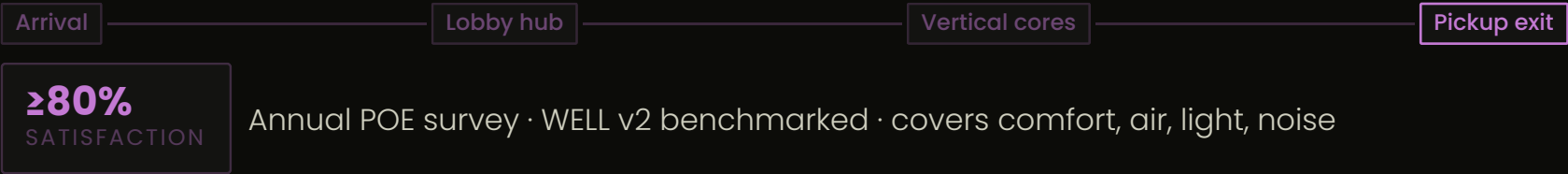
Background noise levels and reverberation by zone



KPI 5.5

People flow + satisfaction

Vertical circulation logic + post-occupancy target



KPI 6.1

Materials hierarchy

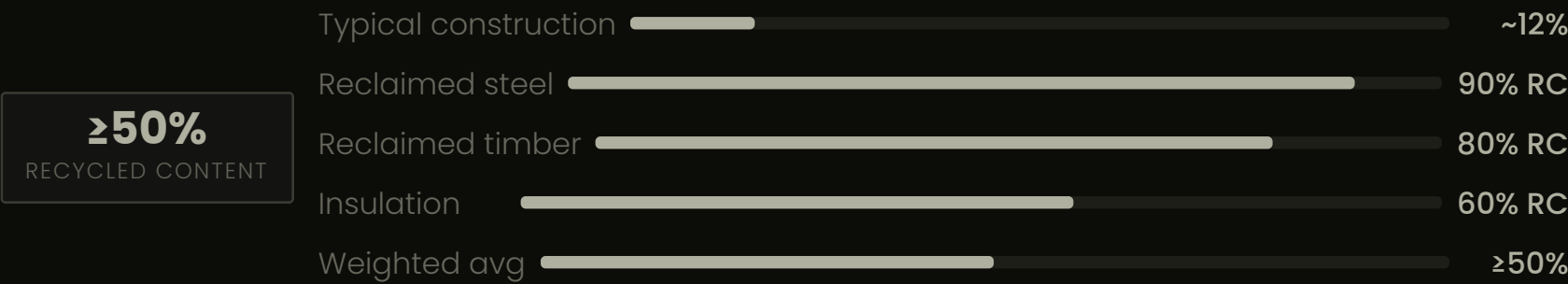
Material selection based on Materialepyramiden — reuse first



KPI 6.2

Recycled material content

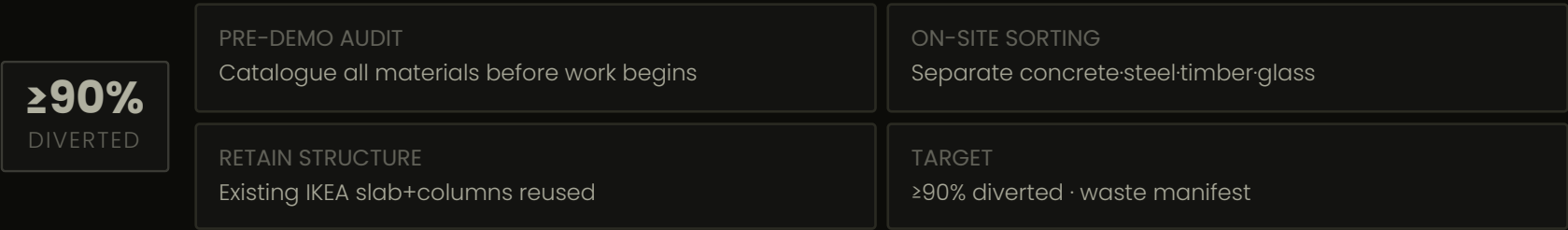
% of total material volume from recycled or reclaimed sources



KPI 6.3

Construction waste diversion

% of construction waste diverted from landfill



KPI 6.4

Material reuse rate

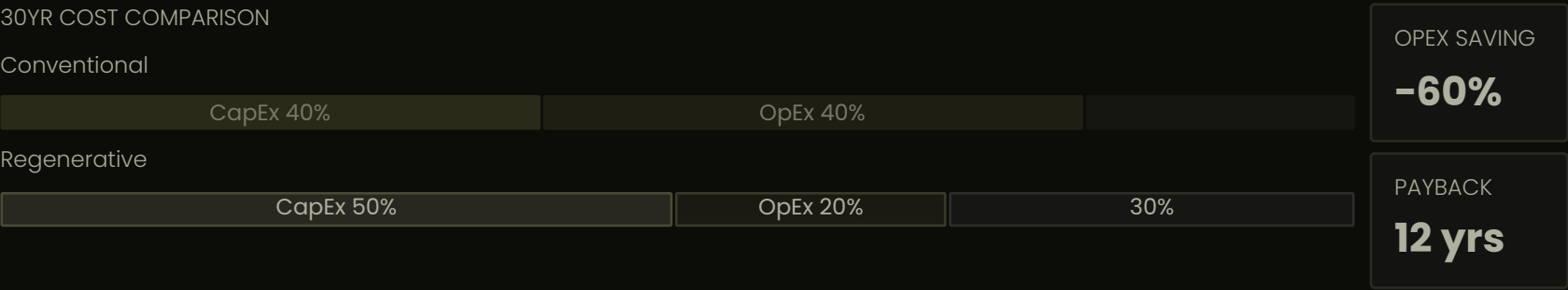
% of existing building fabric retained and reused



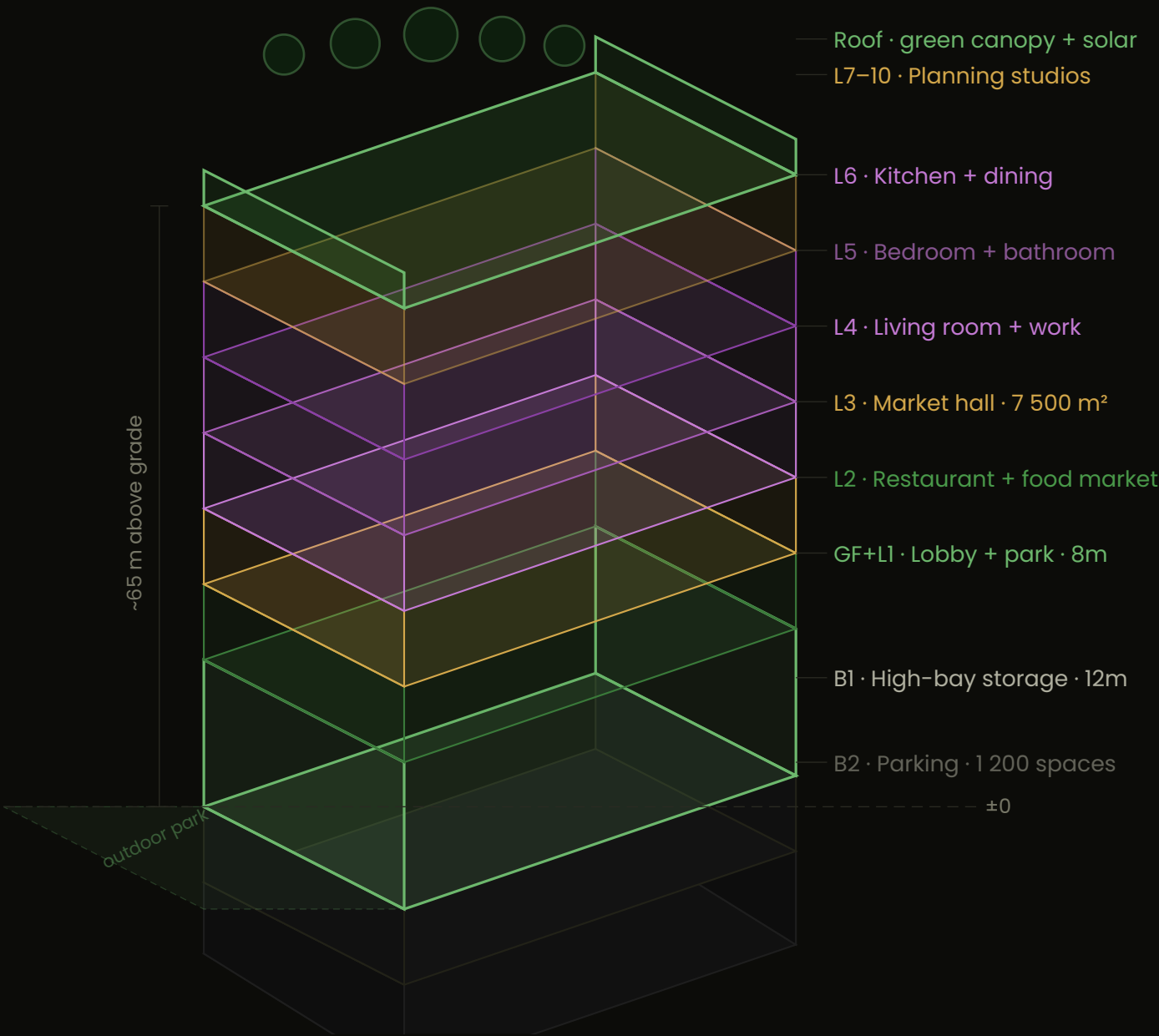
KPI 6.5

Life-cycle cost

30yr total cost: capital + operational + maintenance



■ B2 PARKING + B1 STORAGE ■ GF+L1 LOBBY + PARK ■ L2 RESTAURANT ■ L3 MARKET HALL ■ L4-6 SHOWROOMS ■ L7-10 STUDIOS ■ ROOF GREEN + SOLAR



NEW BUILDING PUBLIC GREEN POND ECO CORRIDOR TRUCK ACCESS PEDESTRIAN + CYCLE CONTEXT





PARAMETRIC SYSTEM -
GRASSHOPPER / RHINO

Three parametric modules generate the building configuration based on the needs of any city site. All dimensions adjustable - the system produces a different building for each location while maintaining the same regenerative logic.

EXTERIOR VISUALISATION

SOUTH-WEST ELEVATION · DUSK · ČERNÝ MOST, PRAHA 14

CASCADING GREEN TERRACES

TIMBER + GLASS HYBRID

D10 HIGHWAY CONTEXT

ROOFTOP FOREST CANOPY

A regenerative approach to urban logistics — integrating storage, retail, showrooms, and public space within a single vertical structure. Each visualisation directly responds to the KPI framework established at project outset.

~65 m
BUILDING HEIGHT

10–14
FLOORS ABOVE GROUND

62%
CANOPY COVERAGE

+35%
BIODIVERSITY NET GAIN

MATERIALS

The lobby is defined by a heavy emphasis on exposed CLT and timber, which serves as the primary structural and aesthetic driver for the interior. This material choice, combined with a central atrium void, creates a sense of openness and connectivity across multiple levels. By utilizing natural wood, the design establishes a warm, “regenerative” atmosphere that directly impacts the user experience.

INTERIOR VISUALISATION — LOBBY

CENTRAL ATRIUM · GROUND FLOOR · PUBLIC NODE · DAYLIGHT VOID

MULTI-LEVEL ATRIUM VOID

EXPOSED CLT + TIMBER

NATURAL DAYLIGHT · KPI 5.1

LIVING WALLS · KPI 1.2

A regenerative approach to urban logistics — integrating storage, retail, showrooms, and public space within a single vertical structure. Each visualisation directly responds to the KPI framework established at project outset.



≥65%
DAYLIGHT AUTONOMY

≤800
PPM CO₂ TARGET

6
HABITAT TYPES INSIDE

≥80%
OCCUPATION SATISFACTION



GREEN CANOPY AND ITS BENEFITS

A vital layer of the structure that transforms a traditional logistics roof into a high-functioning ecological asset. By integrating a bio-diverse meadow and substantial greenery. The rooftop serves as a critical junction where mechanical performance meets biological life, offering several key benefits

ROOFTOP — GREEN CANOPY

ROOF LEVEL · BIODIVERSITY HABITAT · SOLAR INTEGRATION

BIODIVERSE MEADOW · KPI 1.1

62% CANOPY · KPI 1.3

SOLAR PV · KPI 2.4

RAINWATER RETENTION · KPI 3.2

A regenerative approach to urban logistics — integrating storage, retail, showrooms, and public space within a single vertical structure. Each visualisation directly responds to the KPI framework established at project outset.

62%

TREE CANOPY COVERAGE

≥90%

RENEWABLE ENERGY SHARE

320 m³

RAINWATER BUFFER

+35%

BIODIVERSITY NET GAIN

MARKET HALL — L3

FOOD MARKET · FRESH PRODUCE · BISTRO · 7 500 M² · FLOOR 3

EXPOSED CLT + GLULAM · KPI 6.1

HANGING PLANTS · KPI 1.2

SKYLIGHTS · KPI 5.1

NATURAL VENTILATION · KPI 5.2

A regenerative approach to urban logistics — integrating storage, retail, showrooms, and public space within a single vertical structure. Each visualisation directly responds to the KPI framework established at project outset.



7 500

M² MARKET HALL

1 250

M² BISTRO + FOOD

≥65%

DAYLIGHT AUTONOMY

CLT

TIMBER STRUCTURE

VERTICALITY OPENS OPEN
SPACE AND PROVIDES
POSSIBILITY OF CONNECTIONS

Extending space beyond the horizontal plane and creating a stronger connection to the surrounding environment. It allows spaces to interact more naturally with their context feeling more connected, expansive, and integrated rather than enclosed or isolated.

ECOLOGICAL CONNECTIVITY BRIDGE

KPI 1.5 · BIODIVERSITY CORRIDOR · D10 CROSSING · ROKYTKA VALLEY

ECOLOGICAL CORRIDOR · KPI 1.5

NATIVE PLANTING · KPI 1.1

CYCLING · KPI 4.3

TIMBER STRUCTURE · KPI 6.1

A regenerative approach to urban logistics — integrating storage, retail, showrooms, and public space within a single vertical structure. Each visualisation directly responds to the KPI framework established at project outset.

+35%
BIODIVERSITY NET GAIN

6
HABITATS CONNECTED

300 m
CYCLE ROUTE A26

E—W
FULL ECOLOGICAL AXIS

Instagram Social Media Manager

Responsible for creating, designing and publishing all content for the REGENERATIVE.LOGISTICS Instagram account throughout the full semester — from week 1 kickoff through to the final outcome.

- Weekly updates — documenting studio progress
- Speaker posters — announcing guest lectures
- Special posts — hackathon · mid-term · collaborations

33

posts total

18

speaker posters

12

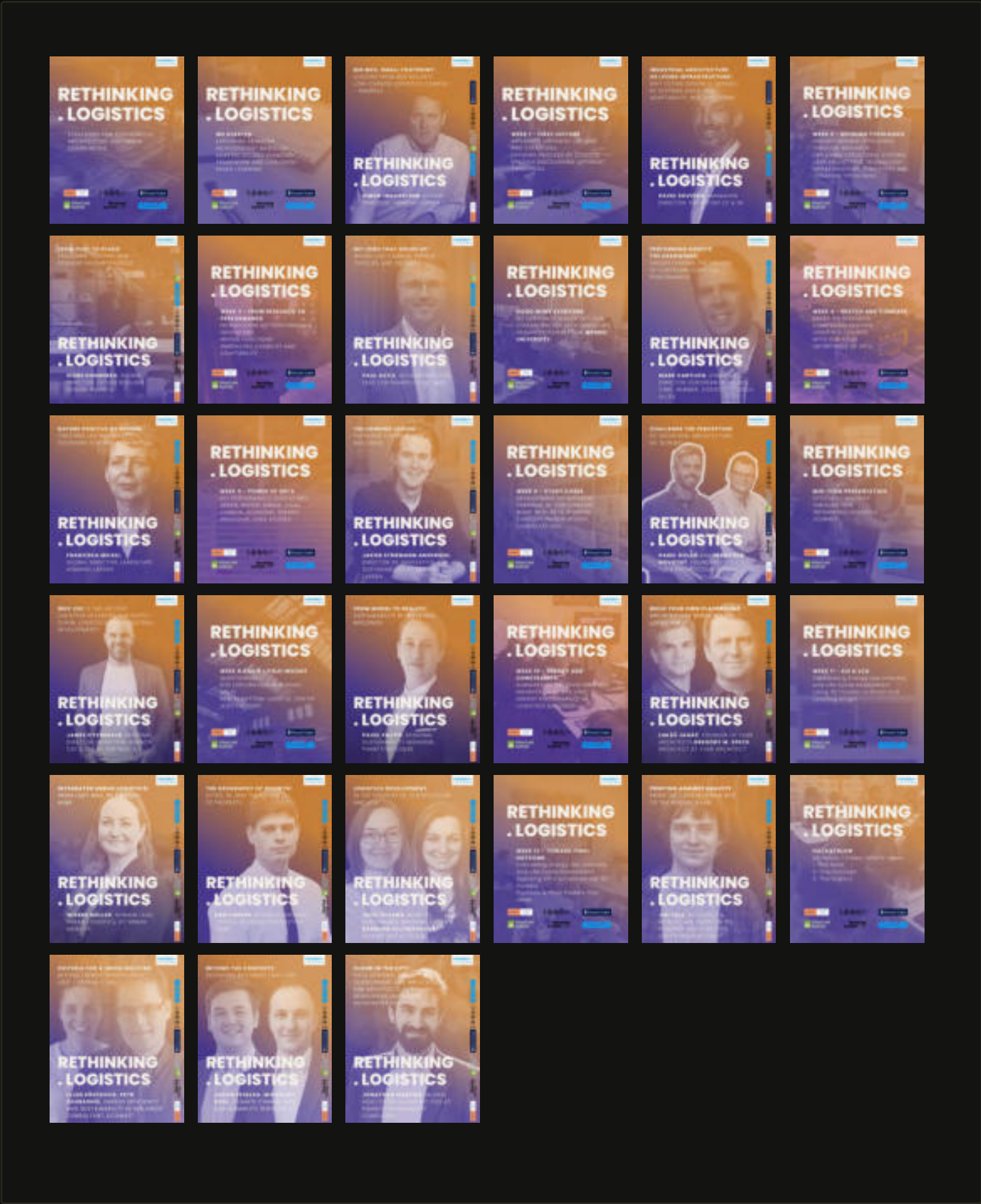
weekly updates

3

special events

Consistent visual identity — purple-orange gradient, bold condensed type, partner logos, Rambøll Fonden branding.

ALL 33 POSTS — CHRONOLOGICAL ORDER 1 → 33



Studio Content Creator

Documenting the full semester through photography, video and interviews — creating a visual archive of the studio's work, process and people for use in teaser films and promotional content.

- Photography — capturing lectures, workshops and studio moments
- Intro movies — helping peers with opening sequences
- Interviews — students + tutors talking about the studio
- Teaser content — raw footage for final films

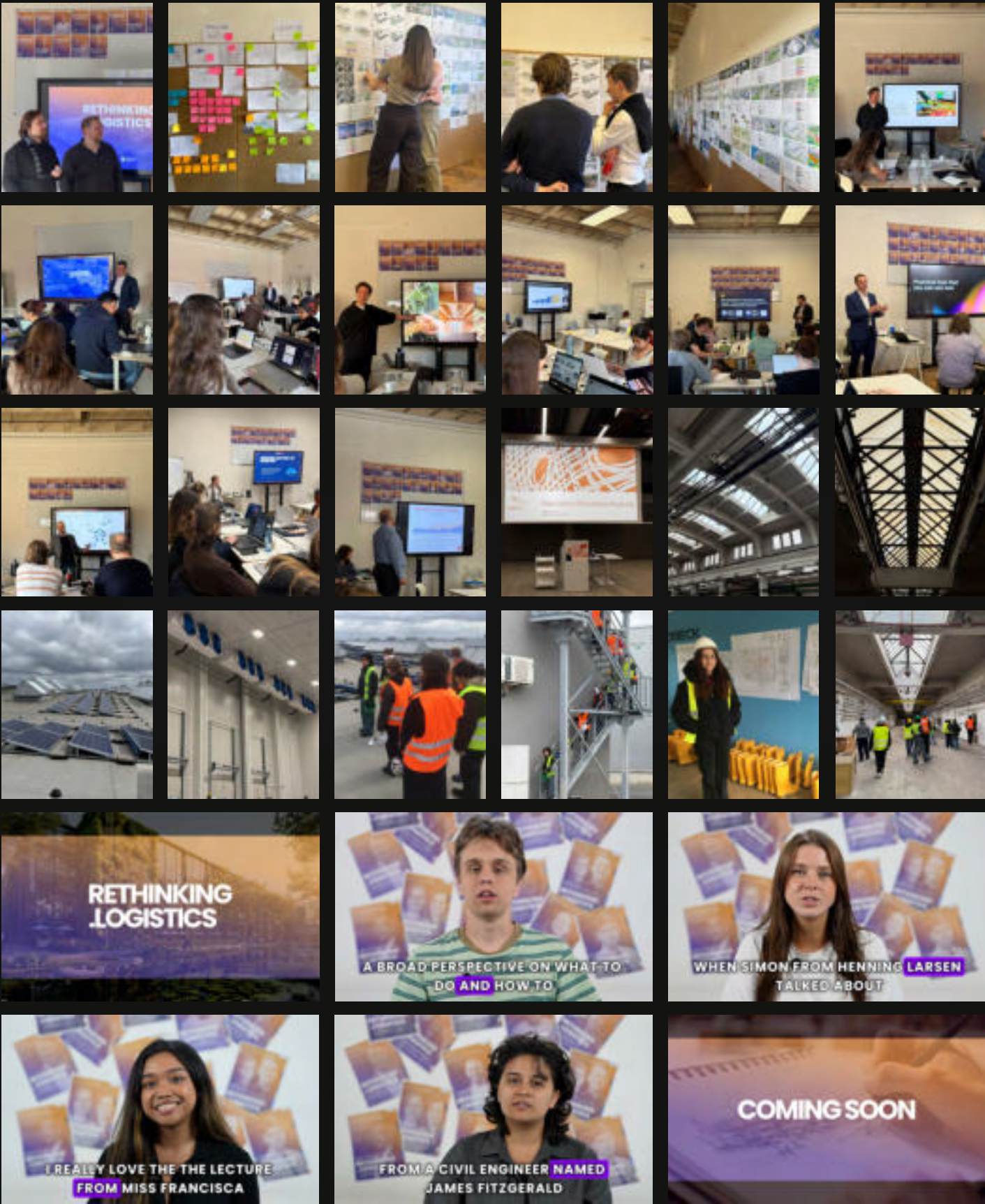
12+

weeks documented

∞

photos taken

PHOTOGRAPHY + VIDEO STILLS



Exhibition Print Outputs

ARCHIP

Portfolio

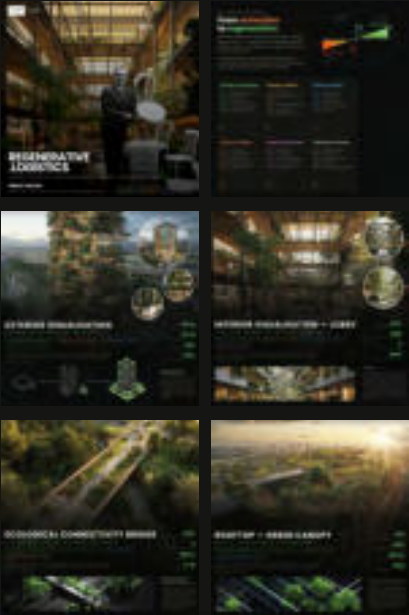
This portfolio entry serves as a comprehensive roadmap of the methodology behind the REGENERATIVE.LOGISTICS project, detailing a workflow that balances technical precision with visionary design. Through a curated collection of conceptual diagrams, plans, and sections, the presentation unfolds the logic of the custom generative scripts used to “grow” the building’s form. This computational core is further enriched by AI-driven visualizations—synthesized from primary research and lecture data—to project a tangible future for industrial space. By anchoring these creative outputs to specific Key Performance Indicators (KPIs), the portfolio demonstrates a rigorous, data-backed approach to architecture, positioning the project as a definitive benchmark for high-performance, Human-Integrated Logistics.



ARCHIP

6 Square Posters

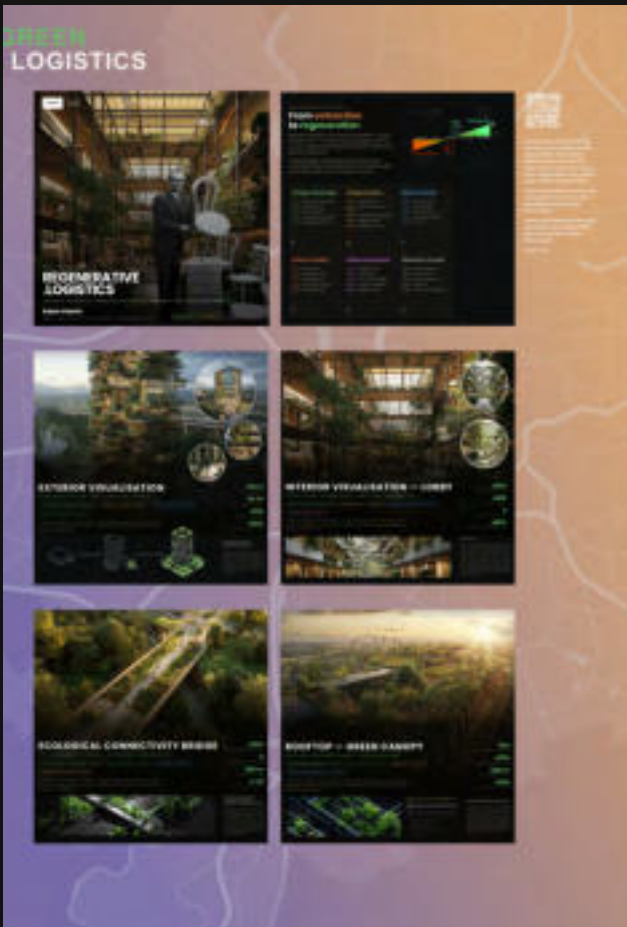
This series of six square posters serves as the visual and analytical core of the project, distilling the complex layers of REGENERATIVE.LOGISTICS into its most vital solutions. Each poster acts as a deep dive into a specific phase of the workflow—moving from the initial conceptual manifesto to the custom generative scripts that defined the building’s form. The series highlights the technical precision of our sectional plans and diagrams, while integrating the high-fidelity AI visualizations derived from our lecture research and data synthesis. By directly linking every architectural intervention to its corresponding Key Performance Indicator (KPI), these posters provide a definitive summary of the project’s success as a high-performance, human-integrated, and truly regenerative logistical prototype.



NTK

Studio Poster

Studio poster that uses the studio Kurilla - Prokop gradient as a background and shows the 6 square posters highlighting the important parts of the project. The poster includes QR code that leads you to the full portfolio to discover the full potential and details of the project.



More graphical and printed outputs to be generated after the final submissions for future events of RETHINKING.LOGISTICS studio as is Round Table, presentation on social media or KPI handbooks for Logistics.



Adam Večeřa

Regenerative.Logistics · Summer 2025–26 · ARCHIP Prague

What I learned

A semester at the intersection of logistics, sustainability, parametric design and communication.

01 Performance before form

KPIs are not a checklist — they are a design generator. Starting from measurable targets and working backwards to spatial decisions is a fundamentally different way of designing.

02 Parametric thinking scales ideas

Writing Grasshopper scripts that generate different building configurations taught me that a good idea should be a system, not a single object.

03 Logistics is urban design

A warehouse is not a background building — it is a node in a city system. Getting that right has enormous consequences for ecology, mobility and community.

04 Communication is part of the work

Managing the Instagram account and creating content for the studio taught me that how you present ideas is inseparable from the ideas themselves.

SEMESTER OUTPUTS

ARCHITECTURAL PROJECT

IKEA Černý Most — vertical regenerative logistics tower

Full portfolio · 32 KPI metrics · site plan · isometric · sections · elevations · MidJourney visualisations · Grasshopper script

SOCIAL MEDIA

REGENERATIVE.LOGISTICS Instagram account

32 posts — 17 speaker posters · 12 weekly updates · consistent visual identity across the full semester

STUDIO CONTENT

Photography · interviews · teaser films

Documenting lectures, workshops and studio process · helping with intro movies · creating interview content for teaser films: The Hook · The Concept · The Impact

POSTERS

A0 format poster series — three posters

Poster A: Project overview · Poster B: Human integrated logistics · Poster C: Green solutions · a unified style · full A0 print-ready

*The **RETHINKING.LOGISTICS** atelier pushed me
to be simultaneously an architect, a data
analyst, a parametric designer, communicator
and content creator.*

Great combination of skills for my future career.

